25900332 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCCFS200 - INTRODUCTION TO PROGRAMMING

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **10** questions

Each question carries **1** marks

1. Describe I/O operator. [An] / [CO2]
2. Estimate the advantage of using a structured programming language. [U] / [CO1]
3. Interpret the meaning of polymorphism. [An] / [CO2]
4. Present the difference between Break and Continue in loops. [An] / [CO2]
5. Describe how can you recognize that a function belongs to a class. [An] / [CO2]
6. Examine how 2-D array can be represented in memory. [An] / [CO2]
7. Compare friend function and a member function. [E] / [CO3]
8. Differentiate between member function and non member function. [E] / [CO3]
9. Mention two characteristics of Constructor. [E] / [CO3]
10. Define parameterised constructor [E] / [CO3]

11. Why is virtual base class is used? [E] / [CO3]

12. Write any two rules of operator overloading. [E] / [CO3]

[1x10 = 10]

Part B

Short Essay Type Questions

Answer any **4** questions

Each question carries **5** marks

13. Classify the different symbols used in Flowchart and explain their functions. [U] / [CO1]

14. Differentiate between testing and debugging [An] / [CO2]

15. Can access specifiers be used more than once in a class? [An] / [CO2]

16. Analyze the effect of modifying a parameter in a function using call by reference. What happens to the original variable? [E] / [CO3]

17. Explain hierarchical inheritance with suitable example and diagram. [E] / [CO3]

18. Explain single inheritance with suitable example and diagram. [E] / [CO3]

[5x4 = 20]

Part C

Essay Type Questions

Answer any **2** questions

Each question carries **10** marks

19. a) What is program structure of C++? b) Illustrate C++ Data Types. [An] / [CO2]

20. Write two C++ programs to swap the values of two numbers: one using call by value and the other using call by reference. Analyze the differences in the outputs and explain why the results differ between the two methods. [E] / [CO3]

21. Define Base class. [E] / [CO3]

[10x2 = 20]